A RATIONAL MAP INTO A PROPER SCHEME CAN BE MADE A MORPHISM BY A MODIFICATION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file divisors.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Carrier window: statement and proof both lie inside the window 8000-10000 granted to this lane by CONTROL 8.83.

Editorial note. The references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: sources/stacks-divisors/retained-preamble.tex (88 lines, verbatim present).

Allowed standard prerequisites (editorial).

- Definition of the scheme theoretic image. Morphisms, Definition morphisms-definition-scheme-theoretic-image, used at proof lines 9464-9468. Constructs X'.
- Properness is stable under base change and composition, and closed immersions are proper. Morphisms, Lemmas morphisms-lemma-base-change-proper, morphisms-lemma-closed-immersion-proper and morphisms-lemma-composition-proper, used at proof lines 9468-9473. Make p : X' -> X proper.
- The scheme theoretic image of a partial section over a retrocompact open. Morphisms, Lemma morphisms-lemma-scheme-theoretic-image-of-partial-section, used at proof lines 9473-9477. Gives p^{-1}(U) = U.

Source reference. Original definition 01R7: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2292–2292. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01W4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2469–2469. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01W5: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2470–2470. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01W3: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2468–2468. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Date: stacks-project:divisors.tex:local-snapshot-20261012; GNU FDL 1.2 or later.

Source reference. Original lemma 0CNG: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 15891–15891. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let X, Y be schemes over S. Assume X is Noetherian and Y is proper over S. Given an S-rational map $f : U \to Y$ from X to Y there exists a morphism $p : X' \to X$ and an S-morphism $f' : X' \to Y$ such that*

(1) *p is proper and $p^{-1}(U) \to U$ is an isomorphism,*
(2) *$f'|_{p^{-1}(U)}$ is equal to $f \circ p|_{p^{-1}(U)}$.*

Proof. Denote $j : U \to X$ the inclusion morphism. Let $X' \subset Y \times_S X$ be the scheme theoretic image of $(f, j) : U \to Y \times_S X$ (Morphisms, Definition 01R7). The projection $g : Y \times_S X \to X$ is proper (Morphisms, Lemma 01W4). The composition $p : X' \to X$ of $X' \to Y \times_S X$ and g is proper (Morphisms, Lemmas 01W5 and 01W3). Since g is separated and $U \subset X$ is retrocompact (as X is Noetherian) we conclude that $p^{-1}(U) \to U$ is an isomorphism by Morphisms, Lemma 0CNG. On the other hand, the composition $f' : X' \to Y$ of $X' \to Y \times_S X$ and the projection $Y \times_S X \to Y$ agrees with f on $p^{-1}(U)$. □